Operating Systems Synchronization - Quick Revision

Critical Section

Definition: The part of a program where shared resources (memory, files, variables, printers, etc.) are accessed.
Rule: Only one thread/process should execute here at a time.
Requirements: Mutual Exclusion, Progress, Bounded Waiting.

Race Condition

Definition: A situation where multiple threads access and modify shared data simultaneously, causing unpredictable or incorrect results.

Mutual Exclusion

Definition: A synchronization property that allows only one thread/process to enter the critical section at a time.

Mutex

Definition: A locking mechanism with ownership. The thread that locks it must unlock it.
Use: Protect shared variables and critical sections.

Semaphore

Definition: A synchronization primitive that uses a counter to control access to one or more shared resources.
Types: Binary (0/1), Counting (0..N).

wait()

Definition: Decrements the semaphore count. If no resource is available, the thread blocks until one becomes free.

signal()

Definition: Increments the semaphore count and wakes one waiting thread/process.

Monitor

Definition: A high-level synchronization construct that automatically provides mutual exclusion and manages locking/unlocking internally (e.g., Java synchronized).

Quick Differences

Mutex: ownership, one thread. Semaphore: no ownership, one/many resources.
Monitor: automatic synchronization.

Interview Tips

Mutex -> shared variable. Counting Semaphore -> printer pool, DB connections, thread pool. Monitor -> safer because locking is automatic.